

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

**COURSE NAME:** Cryptography and Network Security

**COURSE CODE:** CSA51

### COURSE OUTCOME COVERED

**CO1:** Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

**QUESTIONS: 5**

**Total Marks:50**

Annexure A

### CONCEPT MAPPING

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#### **Code of Conduct:**

*I \_Pusalapati Chandu\_(192372119) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:**

*Pchandu*

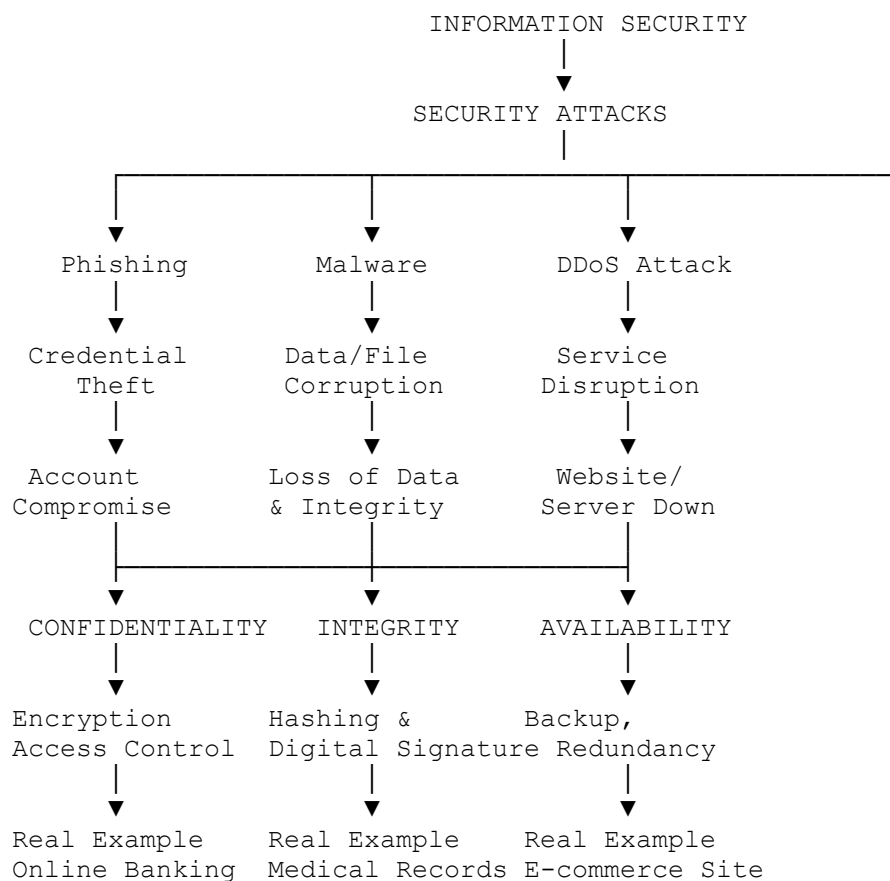
# Question 1: Concept Map – Security Attacks, Threats, Risks and Security Goals

Information security is the process of protecting computer systems, networks, and data from unauthorized access, misuse, modification, or destruction. Organizations face various **security attacks** that create **threats**, which may lead to **risks** affecting valuable information and business operations. To reduce these risks, organizations implement the three primary **security goals** known as the **CIA Triad**:

- **Confidentiality**
- **Integrity**
- **Availability**
- **Authentication**

A concept map helps understand the relationship between attacks, threats, risks, and security goals.

## Concept Map (Diagram)



# Explanation of the Concept Map

## 1. Security Attacks

Security attacks are deliberate attempts to damage, steal, or gain unauthorized access to information systems.

### Types of Attacks

#### Phishing Attack

- Attackers send fake emails or websites to steal login credentials.

#### Malware Attack

- Malicious software such as viruses, worms, spyware, or ransomware infects a computer.

#### DDoS (Distributed Denial of Service) Attack

- Multiple computers flood a server with requests, making the service unavailable.

## 2. Threats

A threat is a possible danger caused by a security attack.

| Attack   | Threat             |
|----------|--------------------|
| Phishing | Credential Theft   |
| Malware  | Data Corruption    |
| DDoS     | Service Disruption |

## 3. Risks

A risk is the possible loss or damage if a threat successfully exploits a vulnerability.

| Threat             | Risk                        |
|--------------------|-----------------------------|
| Credential Theft   | Unauthorized Account Access |
| Data Corruption    | Loss of Data Integrity      |
| Service Disruption | Business Downtime           |

## 4. Security Goals (CIA Triad)

### A. Confidentiality

**Purpose:**

Protects sensitive information from unauthorized access.

**Security Controls**

- Encryption
- Password Authentication
- Multi-Factor Authentication
- Access Control

**Real-World Example**

Online banking encrypts customer passwords and financial information so that only authorized users can access their accounts.

### B. Integrity

**Purpose:**

Ensures that data remains accurate and is not modified without authorization.

**Security Controls**

- Hash Functions
- Digital Signatures
- Checksums
- Version Control

**Real-World Example**

Hospitals use digital signatures to ensure that patient medical records cannot be altered without authorization.

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### C. Availability

**Purpose:**

Ensures that systems and information remain available whenever users need them.

**Security Controls**

- Firewalls
- Backup Servers
- Load Balancing
- Disaster Recovery

**Real-World Example**

An e-commerce website uses backup servers and load balancing so customers can continue shopping even during heavy traffic or cyberattacks.

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# Relationship Between Attacks, Threats, Risks, and Security Goals

| Security Attack | Threat             | Risk               | Security Goal   | Real-World Example                      |
|-----------------|--------------------|--------------------|-----------------|-----------------------------------------|
| Phishing        | Credential Theft   | Account Compromise | Confidentiality | Online banking login protection         |
| Malware         | File Corruption    | Data Loss          | Integrity       | Hospital patient records                |
| DDoS            | Service Disruption | Website Downtime   | Availability    | Amazon or Flipkart website availability |

## Analysis

- Security attacks create threats that exploit vulnerabilities.
- Threats lead to risks that may affect individuals and organizations.
- The CIA Triad helps reduce these risks:
  - **Confidentiality** prevents unauthorized disclosure of data.
  - **Integrity** ensures data accuracy and prevents tampering.
  - **Availability** ensures continuous access to systems and services.
- Applying appropriate security controls significantly reduces the impact of cyberattacks.

## Conclusion

Security attacks such as phishing, malware, and DDoS create different threats and risks to information systems. Organizations mitigate these risks by implementing the **CIA Triad**—**Confidentiality, Integrity, and Availability**—using techniques such as encryption, digital signatures, access control, backups, and disaster recovery. This relationship demonstrates how effective security goals protect information and maintain secure communication systems.

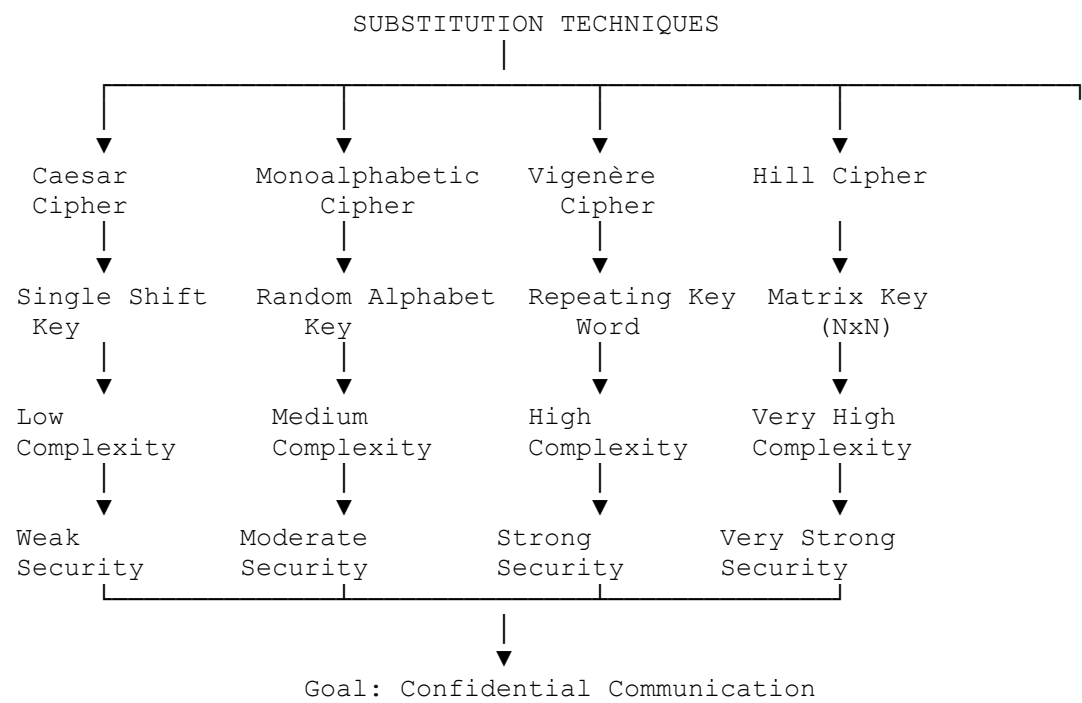
## Question 2: Concept Map – Comparison of Substitution Techniques

Substitution ciphers are classical encryption techniques in which each letter of the plaintext is replaced by another letter or group of letters according to a specific rule or key. These techniques improve confidentiality by hiding the original message from unauthorized users. Different substitution ciphers vary in terms of **key usage, complexity, and security strength**.

The four commonly used substitution techniques are:

- Caesar Cipher
- Monoalphabetic Cipher
- Vigenère Cipher
- Hill Cipher

# Concept Map (Diagram)



## Comparison of Substitution Techniques

| Technique             | Key Usage                        | Complexity | Security Strength |
|-----------------------|----------------------------------|------------|-------------------|
| Caesar Cipher         | Single fixed shift key           | Low        | Weak              |
| Monoalphabetic Cipher | One random substitution alphabet | Medium     | Moderate          |
| Vigenère Cipher       | Repeating keyword                | High       | Strong            |
| Hill Cipher           | Matrix key (2×2, 3×3, etc.)      | Very High  | Very Strong       |

### 1. Caesar Cipher

#### Key Usage

- Uses a **single shift key** (for example, Shift = 3).

#### Complexity

- Very simple algorithm.

#### Security Strength

- Weak because only **25 possible keys** exist.
- Easily broken using brute-force attacks.

#### Example

Plaintext: HELLO

Shift = 3

Ciphertext: **KHOOR**

## 2. Monoalphabetic Cipher

### Key Usage

- Uses one randomly arranged alphabet.

### Example

#### Plain Alphabet

ABCDEFGHIJKLMNOPQRSTUVWXYZ

#### Cipher Alphabet

QWERTYUIOPASDFGHJKLZXCVBNM

### Complexity

- More complex than Caesar Cipher.

### Security Strength

- Stronger than Caesar.
- Still vulnerable to frequency analysis.

## 3. Vigenère Cipher

### Key Usage

- Uses a **repeating keyword**.

### Example

#### Plaintext

HELLO

#### Keyword

KEYKE

### Complexity

- High because every letter uses a different shift.

### Security Strength

- Stronger than Monoalphabetic Cipher.
- Difficult to break without knowing the keyword.

## 4. Hill Cipher

### Key Usage

- Uses a matrix as the encryption key.

### Example

#### Key Matrix

```
| 3 3 |  
| 2 5 |
```

### Complexity

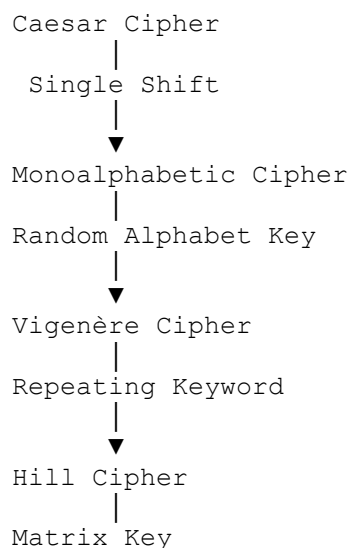
- Very high because matrix multiplication and modular arithmetic are used.

### Security Strength

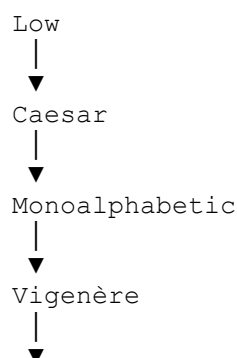
- Strongest among the four classical substitution ciphers.
- Encrypts blocks of letters instead of one letter at a time.

## Relationship Based on Key Usage

### Key Usage

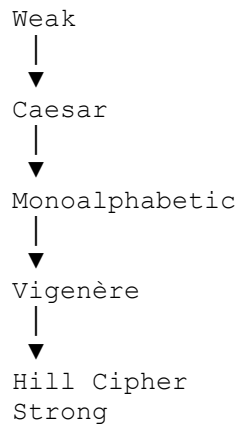


## Relationship Based on Complexity





## Relationship Based on Security Strength



## Analysis: Which Technique is Most Secure?

Among these four techniques, the **Hill Cipher** is the **most secure** because:

- It uses **matrix keys**, making encryption more complex.
- It encrypts **blocks of letters** rather than individual letters.
- Frequency analysis becomes much more difficult.
- It combines linear algebra and modular arithmetic, increasing resistance to classical attacks.

However, compared with modern encryption algorithms such as **AES**, even the Hill Cipher is not considered secure enough for today's communication systems.

## Advantages of Each Technique

### Caesar Cipher

- Easy to learn and implement.
- Suitable for beginners.

### Monoalphabetic Cipher

- More secure than Caesar.
- Uses a larger key space.

### Vigenère Cipher

- Uses multiple alphabets.
- Stronger against frequency analysis.

### Hill Cipher

- Encrypts multiple letters at once.
- Highest security among classical substitution ciphers.

# Conclusion

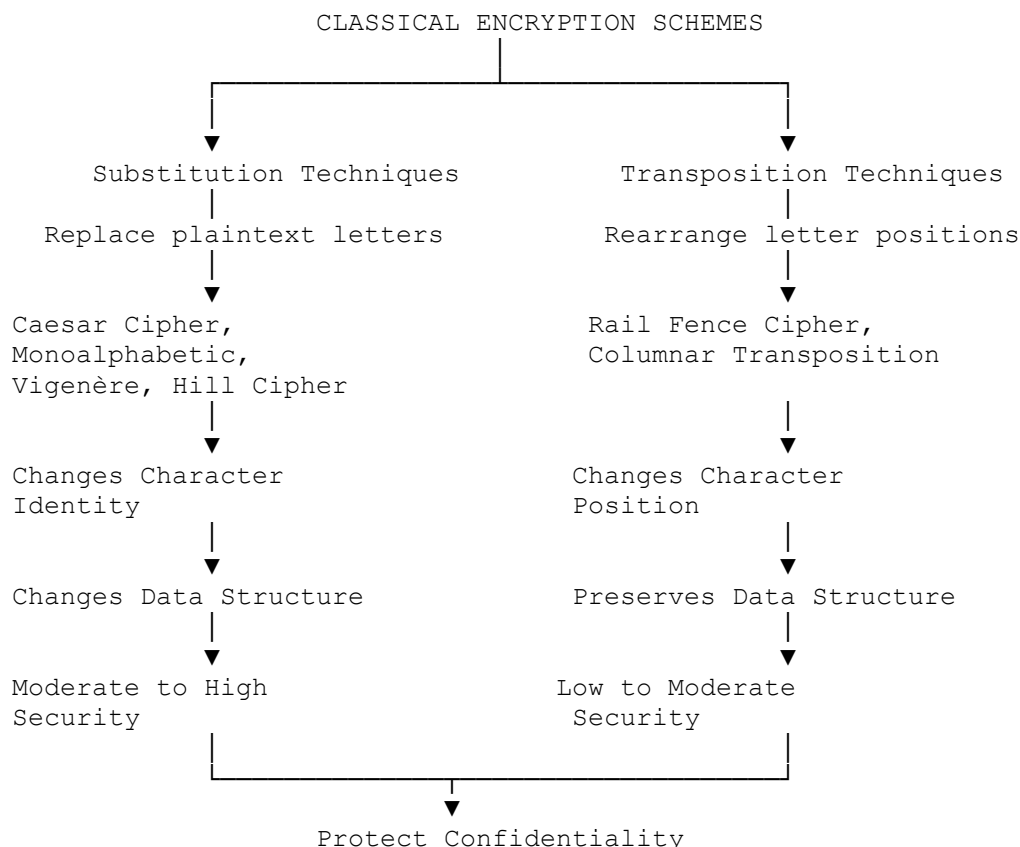
The comparison of **Caesar Cipher**, **Monoalphabetic Cipher**, **Vigenère Cipher**, and **Hill Cipher** shows that security increases as the complexity of the key and encryption process increases. The **Caesar Cipher** is the weakest because it uses only a single shift key, while the **Hill Cipher** is the strongest because it uses matrix-based encryption and block processing. Therefore, among the given classical substitution techniques, the **Hill Cipher provides the highest level of security** due to its mathematical complexity and resistance to classical cryptanalysis.

## Question 3: Concept Map – Classical Encryption Schemes (Substitution Techniques and Transposition Techniques)

Classical encryption schemes are traditional methods of securing information before the development of modern cryptography. They are mainly divided into two categories:

1. **Substitution Techniques** – Replace each plaintext letter with another letter or symbol.
2. **Transposition Techniques** – Rearrange the positions of the plaintext letters without changing the letters themselves.

### Concept Map (Diagram)



# Explanation of the Concept Map

## 1. Classical Encryption Schemes

Classical encryption methods protect information by transforming plaintext into ciphertext. They are classified into substitution and transposition techniques.

## 2. Substitution Techniques

### Definition

Substitution techniques replace each character in the plaintext with another character according to a predefined key.

### Examples

- Caesar Cipher
- Monoalphabetic Cipher
- Vigenère Cipher
- Hill Cipher

### Data Structure

- Changes the identity of characters.
- Original letters are replaced with different letters.

### Example

Plaintext:

HELLO

Ciphertext (Caesar Cipher):

KHOOR

The letters change completely.

## 3. Transposition Techniques

### Definition

Transposition techniques do not replace characters.

Instead, they rearrange the positions of characters.

## Examples

- Rail Fence Cipher
- Columnar Transposition Cipher

## Data Structure

The same letters remain present.

Only their positions change.

Example

Plaintext

HELLO

Possible Transposition

LEHLO

The letters remain the same but appear in a different order.

## Similarities

### Substitution Techniques

Encrypt plaintext

Require an encryption key

Produce ciphertext

Aim to provide confidentiality

Used in classical cryptography

### Transposition Techniques

Encrypt plaintext

Require an encryption key

Produce ciphertext

Aim to provide confidentiality

Used in classical cryptography

## Differences

| Feature            | Substitution Techniques | Transposition Techniques |
|--------------------|-------------------------|--------------------------|
| Working Method     | Replace characters      | Rearrange characters     |
| Character Identity | Changed                 | Not changed              |
| Character Position | Usually remains         | Changes                  |
| Data Structure     | Modified                | Preserved                |
| Examples           | Caesar, Vigenère, Hill  | Rail Fence, Columnar     |
| Security           | Higher                  | Lower                    |

# (i) Effect on Data Structure and Security Level

## Substitution Techniques

### Effect on Data Structure

- Character identities are changed.
- Makes the original message more difficult to recognize.

### Security Level

- Medium to High.
- Depends on the algorithm.
- Hill Cipher provides the highest security among classical substitution techniques.

## Transposition Techniques

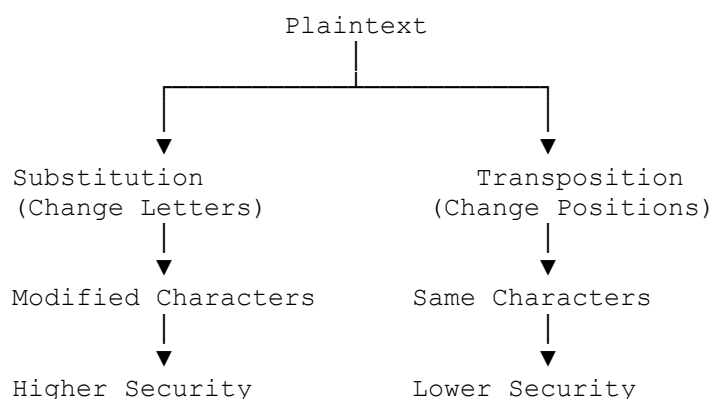
### Effect on Data Structure

- Character positions change.
- Character identities remain unchanged.

### Security Level

- Low to Moderate.
- Easier to analyze because letter frequencies remain unchanged.

## Relationship Diagram



## (ii) Analysis: Which Technique is More Vulnerable and Why?

Transposition techniques are generally more vulnerable to attacks than substitution techniques.

## Reasons

1. The original letters remain unchanged.
2. Letter frequency is preserved.
3. Statistical analysis can easily identify patterns.
4. Brute-force attacks can recover the original arrangement quickly.

## Why Substitution is Better

- Character identities are changed.
- Frequency analysis becomes more difficult.
- Advanced substitution methods such as the Hill Cipher provide better protection.

# Real-World Examples

## Substitution Technique

Military communication in historical times used substitution ciphers to hide sensitive messages.

## Transposition Technique

Early diplomatic messages used Rail Fence and Columnar Transposition ciphers for secret communication.

## Conclusion

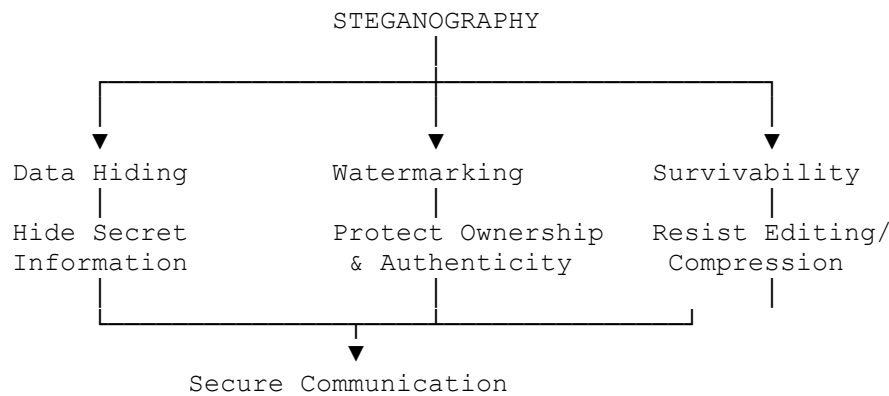
Classical encryption schemes are divided into **Substitution Techniques** and **Transposition Techniques**. Substitution techniques improve security by replacing characters with different symbols, while transposition techniques only rearrange character positions. Although both provide confidentiality, **transposition techniques are more vulnerable** because they preserve letter frequencies and are easier to break using statistical analysis. Therefore, **substitution techniques—especially the Hill Cipher—offer a higher level of security among classical encryption methods.**

## Question 4: Concept Map – Steganography Concepts

### Introduction

Steganography is the technique of hiding secret information inside digital media such as images, audio, or videos. It helps protect confidential information by concealing the existence of the message.

## Concept Map (Diagram)



### (a) Relationship

- **Data Hiding** hides confidential information inside a digital file.
- **Watermarking** protects ownership and authenticity of digital content.
- **Survivability** ensures that hidden data remains intact even after compression or editing.

### (b) Contribution to Secure Communication

- **Data Hiding** keeps the message invisible to attackers.
- **Watermarking** prevents unauthorized copying and proves ownership.
- **Survivability** allows hidden information to remain available after file modifications.

### (c) Real-World Applications

| Concept       | Real-World Application                             |
|---------------|----------------------------------------------------|
| Data Hiding   | Secret military communication                      |
| Watermarking  | Copyright protection for digital images and videos |
| Survivability | Secure sharing of images on social media           |

## Conclusion

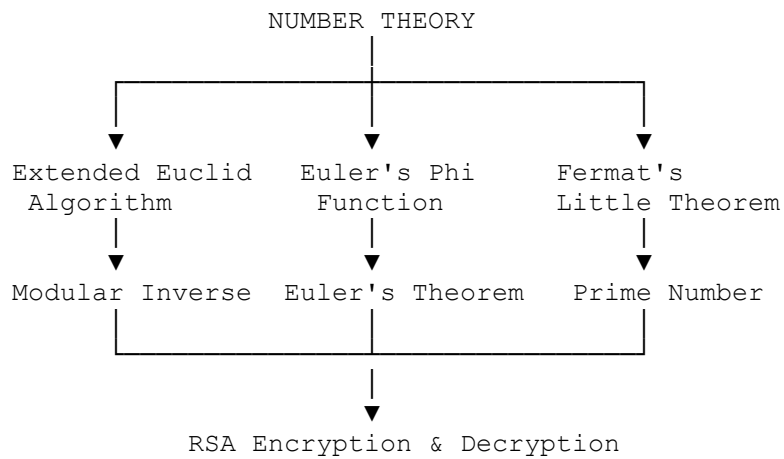
Steganography improves secure communication by hiding confidential data, protecting digital ownership, and ensuring that hidden information survives editing or compression.

## Question 5: Concept Map – Number Theory in Cryptography

### Introduction

Number theory provides the mathematical foundation for modern cryptography, especially public-key algorithms such as RSA.

## Concept Map (Diagram)



### (a) Mathematical Connection

- **Extended Euclid Algorithm** computes the **Modular Inverse**.
- **Euler's Phi Function** is used in **Euler's Theorem**.
- **Fermat's Little Theorem** is a special case of Euler's Theorem for prime numbers.

### (b) Role in Encryption/Decryption

| Concept                   | Role                                   |
|---------------------------|----------------------------------------|
| Modular Inverse           | Calculates private keys                |
| Extended Euclid Algorithm | Finds modular inverse                  |
| Fermat's Little Theorem   | Simplifies modular arithmetic          |
| Euler's Phi Function      | Generates RSA keys                     |
| Euler's Theorem           | Supports RSA encryption and decryption |

### (c) Most Critical Concept

**Euler's Phi Function** is the most important because it is used to generate RSA public and private keys. Without the Phi function, RSA key generation is not possible.

## Conclusion

Number theory concepts work together to make public-key cryptography secure. Among them, **Euler's Phi Function** plays the most important role because it is essential for RSA key generation and secure encryption/decryption.